Title

Survival and harvest rates remain consistent for wood ducks following an increased bag limit

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Abstract

In the early 1900s, wood duck *Aix sponsa* populations experienced severe declines due to unregulated hunting and habitat destruction. Federal protection under the Migratory Bird Treaty Act of 1918 and the implementation of nest box programs facilitated their recovery such that sport harvest resumed in 1941. In 2008, the bag limit for wood ducks during the general duck season increased from two to three daily, a change projected to increase the harvest rates. However, estimating harvest and survival rates of adult females remains challenging due to the difficultly of capturing them during preseason banding operations. We analyzed data from 2001–2014 from box-nesting female wood ducks in Louisiana using a live-dead model to estimate survival and calculate harvest rates. Our results indicate the bag limit increase had minimal impact on the survival and harvest rates of box-nesting females. The mean annual survival rate was 68%, with no changes after the increase in bag limit in 2008. Harvest rates also remained consistent across the study period, averaging 3%. Our findings suggest the increase in bag limit has not adversely affected survival of box-nesting wood ducks in Louisiana, allowing for additional hunting opportunities without compromising conservation goals for adult female wood ducks in the state.\

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Introduction

In the early 1900s, wood duck *Aix sponsa* populations experienced a significant decline, largely due to unregulated hunting and the destruction of breeding habitat (Bellrose and Holm 1994). During this era, wood ducks were highly sought by hunters, especially during spring hunts, and their meat was considered a culinary delicacy. Additionally, extensive clearing of forests led to the loss of vital nesting sites as wood ducks are obligate cavity nesters. In 1918, the Migratory Bird Treaty Act made it illegal to hunt wood ducks and provided them with federal protection. This action, combined with implementation of nest box programs to supplement nest sites across their range, was vital to facilitate population recovery. By 1941, wood duck populations had rebounded to the extent that regulated harvesting became feasible, and nest box programs became common.

Today, state wildlife agencies, including the Louisiana Department of Wildlife and Fisheries (hereafter, LDWF), actively maintain and manage wood duck nest box programs. For example, since 1992, LDWF has overseen approximately 2,000 nest boxes across the state to bolster local breeding populations and enhance hunting opportunities (L.A. Reynolds, LDWF, personal communication). Within these programs, nest boxes are equipped with predator guards to both safeguard eggs from nest predators and reduce female mortality. Recognizing the critical role of reproduction in wood duck population dynamics, ensuring survival of females is critical for these programs to achieve their intended goals (Hepp et al. 2020). However, wood ducks are

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an economically- and culturally-important game species, so there is a need to balance conservation efforts with providing harvest opportunity.

When wood duck hunting resumed in 1941, states allowed a harvest of one bird daily during the open hunting season until 1962, when hunters were allowed to take two birds daily (Bellrose and Holm 1994; Garrettson 2007). In 2007, a harvest assessment indicated that wood duck populations could tolerate an increase in the bag limit (Garrettson 2007; Balkcom et al. 2010). Consequently, since 2008, hunters have been permitted to harvest three wood ducks daily during the general duck season (United States Fish and Wildlife Service 2008). The impact of increased harvest on adult females harvest rates remains uncertain, as this age-sex cohort is difficult to capture during preseason banding operations (Garrettson and Howard 2023; Howard et al. 2024). If higher adult female harvest rates result in lower survival, it could negatively affect populations and counteract the effectiveness of nest box programs (Hepp et al. 2020). This concern is diagnosable, because female wood ducks typically return to their natal areas to nest, enabling the marking of individuals and their recapture in subsequent years. Here, we assess the most recent increase in the bag limit (i.e., three bird daily bag limit beginning in 2008) by comparing survival and harvest rates for box-nesting wood ducks in Louisiana before and after the change.

Study Area

Nest boxes were installed state-wide across Louisiana in Wildlife Management Areas, private wetlands, state parks, and U.S. Army Corps of Engineers properties (Figure 1). Most nest box populations were located within the Mississippi Alluvial Valley, with few located in western Louisiana. Preference was given to locations with shallow wetlands

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and emergent vegetation to provide brood concealment. Some nest boxes were also placed in habitats such as cypress swamps, residential areas, lakes, reservoirs, and coastal marshes. Nest boxes were positioned on dry land, the edges of waterbodies, or over water. The bottoms of nest boxes were approximately 1.5–3.0 m above the ground or water surface. Managers accessed nest boxes using ladders, boats, or from the ground. All boxes were built according to the dimensions recommended by Bellrose and Holm (1994), arranged as duplexes (i.e., two boxes mounted to the pole), and mounted on metal poles with baffle cone predator guards.

Throughout the breeding season (January ─ June), low and high temperatures averaged 14.4 and 23.9ºC, respectively (United States Climate Data 2025). Outside the breeding season (July ─ December), mean low and high temperatures were 17.8 and 26.7ºC, respectively. Average precipitation is 13.5 cm during the breeding season and 12.9 cm during the non-breeding period. The landscape of Louisiana is made up of forested uplands, alluvial plains, coastal marshes, prairies, and bluffs (Holcomb et al. 2015). The highest elevation in Louisiana is 163 m, with much of the lower half of the state being near or below sea level. The northwest and western portions of the state are dominated by mixed hardwood-pine stands, and much of the northeastern and central portions are fragmented by agriculture and bottom land hardwoods. The southern portion of Louisiana is dominated by agriculture, remnant prairie, and coastal marsh.

Methods

Nest boxes were routinely inspected across Louisiana at least twice a year: first in January at the onset of the breeding season and again in April when females began renesting (Baldassarre 2014). In addition to these scheduled checks, nest boxes were

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also opportunistically examined between January and July. During checks, damaged nest boxes and predator guards were either repaired or replaced. Females were captured from nest boxes and banded with aluminum United States Geological Survey leg bands.

We compiled Louisiana data from 2001–2014 for wood ducks that were banded from nests boxes and subsequently harvested by hunters or recaptured from nest boxes. First, we obtained all wood duck band data for our sampling period through the USGS Bird Banding Laboratory (hereafter, BBL) and filtered it to include only individuals that were captured and banded from an artificial nest structure (i.e., EXTRA_INFO_CODE = 33); we classified all individual ages as after-hatch-year. However, the BBL possessed few of these records and so we obtained paper records of females that were recaptured from nest boxes from LDWF and entered these data manually. To identify individuals harvested by hunters, we filtered the encounter column to include only those recorded as being "shot" (E_HOW_OBTAINED_CODE = 1). We retained individuals that were harvested both within and beyond Louisiana state borders (Figure 1). Our final dataset included 6,182 female wood ducks banded from nest boxes, of which 473 were harvested by hunters; 147 were direct recoveries (i.e., harvested during the hunting season immediately following band deployment; Figure 1). During the same years, 1,137 recaptures were made during subsequent breeding seasons by LDWF personnel conducting nest box checks (Figure 1).

We analyzed these data using a Bayesian live-dead model (hereafter, LD model; Burnham 1993, Lebreton et al. 1999, Kéry and Schaub 2012; Figure 2). The LD model provides estimates of survival (S), fidelity (F), recapture (p), and reported mortality (r;

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Kéry and Schaub 2012). The true survival probability, *S*, accounts for emigration outside the sampling area. The probability that an individual remains faithful to the banding area is represented by *F*. The probability of recapturing an individual from a nest box, given it is alive and within the study area, is denoted by *p*. Finally, *r* represents the probability that a dead individual is found and reported. Estimates from LD models remain unbiased when *p* and *r* are ≥0.20 and ≥0.10, respectively (Weegman et al. 2020).

We executed the LD model using a multi-state framework, where female *i* took on one of the following four states: alive in study area, alive outside study area, recently dead, and dead (Figure 2; Data S1; Kéry and Schaub 2012). These comprised the state transition matrix (Ω):

$$z_{i,t+1}|z_{i,t} \sim categorical(\Omega_{zi,t,1\ldots s,i,t}),$$

where the true state of individual *i* at time *t* was $z_{i,t}$. We used the categorical distribution to predict the probability that individual *i* was in state *s* at time $t + 1$ ($z_{i,t+1}$) given its current state ($z_{i,t}$). We linked the state transition matrix to field observations, where individual *i* was able to be detected in one of the following observational states (*o*): seen alive, recovered dead, and not seen or recovered (Figure 2; Data S1). These states made up the observation matrix (θ):

$$y_{i,t}|z_{i,t} \sim categorical(\theta_{zi,t,1\ldots o,i,t}),$$

where the observed state of individual *i* at time *t* was $y_{i,t}$, given its current true state ($z_{i,t}$). We used our observation matrix to estimate *p* using the categorical distribution. We did not include *r* in θ matrix but did include Ω, following updated model guidelines (Kéry and Schaub 2012; Weegman et al. 2020). Important to the framework we used is that only

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recently dead individuals can be recovered as dead. Once an individual entered the dead state, it remained there for all subsequent years.

We fit the LD model in JAGS (Plummer 2003) using the R2jags (Su and Yajima 2021) package in Program R 4.2.2 (R Core Team 2023). We considered annual inconsistencies in sampling efforts by including year as a random effect in the model. We used vague uniform (0,1) priors for *S*, *F*, *p*, and *r* (Kéry and Schaub 2012; Hobbs and Hooten 2015). We used the normal distribution with a mean of 0 and variance (σ^2) to obtain estimates of annual variation. We ran four chains of 50,000 iterations, after considering 25,000 as burn-in, and saved every twentieth iteration. We confirmed our models converged using the Gelman-Rubin statistic, where $\hat{R}$ values < 1.05 indicated convergence (Gelman and Rubin 1992). We reported posterior means and 95% credible intervals (hereafter, CRI) for the LD model. We used posterior means and standard deviations to calculate the coefficient of variation (hereafter, CV) for each parameter to provide measures of precision.

We calculated annual harvest rates on box-nesting wood ducks by dividing the count of direct recoveries (i.e., total individuals marked the breeding season prior and harvested during the proceeding hunting season) by the total number of banded individuals. We then divided this result by annual band reporting rates to account for bands that were recovered by hunters but not reported (Garrettson et al. 2014; Garrettson 2024). For harvest rates, we reported frequentist means and 95% confidence intervals (hereafter, CI). We used credible intervals to assess whether parameters changed following the bag limit increase, with overlapping credible intervals

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183 indicating no significant difference. We conducted our work under USGS banding permit
184 #06669 and Special Use Permit 43614-20-04.

185 **Results**

186 Banders marked an average of 442 females (range = 257— 570) from nest boxes each
187 year. During the hunting period, hunters harvested an average of 34 individuals (range
188 4— 54) each year. A total of 29 harvested individuals were recovered outside Louisiana
189 state boundaries (Figure 1). Banders recaptured 106 females (1— 366) from nest boxes
190 each year on average. Most individuals were captured on a single occasion; however,
191 some females were captured across seven different breeding seasons.

192 For the entire study period, mean survival probability was 0.684 (CRI = 0.618,
193 0.951), mean fidelity probability was 0.735 (CRI = 0.488, 0.970), mean recapture
194 probability was 0.281 (CRI = 0.156, 0.415), and mean reported mortality probability was
195 0.101 (CRI = 0.067, 0.379). The annual variation in these probabilities was substantial,
196 where the CV for survival was 0.116, 0.140 for fidelity, 0.239 for recapture, and 0.845 for
197 reported mortality. Estimates of all four parameters were similar following the increase in
198 daily bag limit (Figure 3); however, recapture probability was lower in 2009. The mean
199 harvest rate was 0.032 (CI = 0.027, 0.038) during the study period and ranged from
200 0.012− 0.038. The average annual harvest rate prior to 2008 was 0.033 (CI = 0.014,
201 0.051) and 0.029 (CI = 0.01, 0.049) after the bag limit increase (Figure 4).

202 **Discussion**

203 Our results suggest that increasing the daily bag limit from two to three birds had a
204 minimal effect on the survival of female box-nesting wood ducks. Likewise, fidelity,

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recapture, and reported mortality probabilities were similar before and after the bag limit increase. Annual harvest rates were comparable before and after the bag limit increase, suggesting hunting related mortality for box-nesting females did not increase. Previous studies conducted in the southeastern United States have produced apparent survival estimates for box-nesting wood ducks (Hepp et al. 1987; Dugger et al. 1999). Apparent survival estimates tend to be biased low because they do not allow for emigration out of the study area. Despite this limitation, apparent survival was 55% for box-nesting wood ducks in South Carolina (Hepp et al. 1987) and 63% in Missouri (Dugger et al. 1999), which closely resembled the mean survival probability (68%) of our study population. The similarity between apparent and true survival estimates is likely due to the strong fidelity exhibited by females to their nest box populations, where only a small portion of banded females leave their nest box populations once they attempt to breed there. We observed high probabilities of fidelity, which averaged 74%. These findings align with Hepp and Kennamer (1992), who reported that 80% of adult females remained faithful to their nest box population.

Our results do not agree with one recent study from Ohio that found adult and juvenile female survival decreased by 8% and 4% following the increase in bag limit (Shirkey and Gates 2020). Likewise, our findings contrast with Trost (1988) who found survival was lower during years when harvest rates were high in the southeastern United States. One major difference between our analysis and these from the literature is that we examined data collected during the breeding season for wood ducks marked from nest boxes, while others have used preseason band data (Trost 1988; Garrettson 2007; Shirkey and Gates 2020). Our estimates of annual survival for box-nesting

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228 females (68%) were higher than estimates from adult females banded preseason in the
229 same geographic region and temporal period (50% and 51%; Garrettson and Howard
230 2023; Howard et al. 2024). Our results may have differed if we had considered
231 Louisiana preseason band data, as such a dataset encompasses individuals that nest in
232 natural cavities, which might result in lower survival probabilities during the breeding
233 season when compared to nest boxes equipped with predator guards. Therefore, our
234 results are only applicable to the box-nesting wood ducks.

235 Harvest rates remained consistent over the study period and were lower than
236 estimates for adult female wood ducks in Ohio (7%; Shirkey and Gates 2020) and the
237 Atlantic flyway (7%; Zimmerman et al. 2017). The lower harvest rate estimates from our
238 study can be explained by findings from several other studies. First, adult female wood
239 ducks are less vulnerable to harvest compared to other sex-age cohorts (Garrettson
240 2007), which may explain why survival and harvest rates were unaffected by the bag
241 limit increase. Secondly, our study population consisted of southern latitude breeding
242 females (Bellrose and Holm 1994; Baldassarre 2014), which allows them to forgo
243 migration, and inhabit areas they are familiar with year-round, thus potentially more
244 effectively avoiding harvest (Bowers and Martin 1975). Next, Balkcom et al. (2010)
245 found the greatest change in harvest rate would be observed when increasing the bag
246 limit from one to two birds, which occurred in 1962, while increasing the limit to three
247 birds daily would result in a lesser change in harvest rate. Lastly, in Fallon, Nevada
248 harvest rates remained similar during experimental hunting seasons, when the daily bag
249 limit was increased to seven birds (Sedinger et al. 2019), providing further evidence that
250 even when bag limits are increased, harvest rates do not always change.

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Given Louisiana is a wintering area for wood ducks migrating from farther north, future research may focus on identifying the major natal regions of wood ducks harvested by hunters. It is plausible that wood ducks hatched outside Louisiana are the primary source of birds harvested by hunters. Knowing such information would allow managers to allocate additional efforts to conserve these populations. For local breeding box-nesting wood ducks, our results suggest the increase in bag limit has provided additional harvest opportunities, without altering the survival of these populations.

Supplemental Materials

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Data S1. Encounter history used in a Bayesian live-dead model for female wood ducks (*Aix sponsa*) captured and marked from nest boxes in Louisiana from 2001–2014. Each row represents an individual, with values ranging from 1 to 3: “1” indicates the individual was observed alive, “2” indicates the individual was harvested by a hunter, and “3” indicates the individual was not observed. Each column represents a year, starting in 2001 and ending in 2014. The live-dead survival model was used to derive survival probabilities from 2001–2014 to investigate the consequences of increasing the daily wood duck bag limit from two to three birds in 2008.

Reference S1. Garrettson PR. 2007. A proposed assessment and decision making framework to inform harvest management of wood ducks in the Atlantic and Mississippi

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Reference S2. Garrettson PR. 2024. Direct recovery, harvest, and kill rates of wood ducks in North America 2003-2022. United States Fish and Wildlife Service, Washington D.C., USA

Reference S3. Howard C, Keever A, Greenwalt A, Garrettson P, Hagy H, Cohen B. 2024. Banding needs assessment for wood duck population monitoring in the Mississippi and Atlantic flyways. United States Fish and Wildlife Service, Washington D.C., USA.

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obtained additional grants; DLB and KMR conceived and conducted the analyses, DLB drafted the paper with edits from KMR, LAR, and JO.

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Figures

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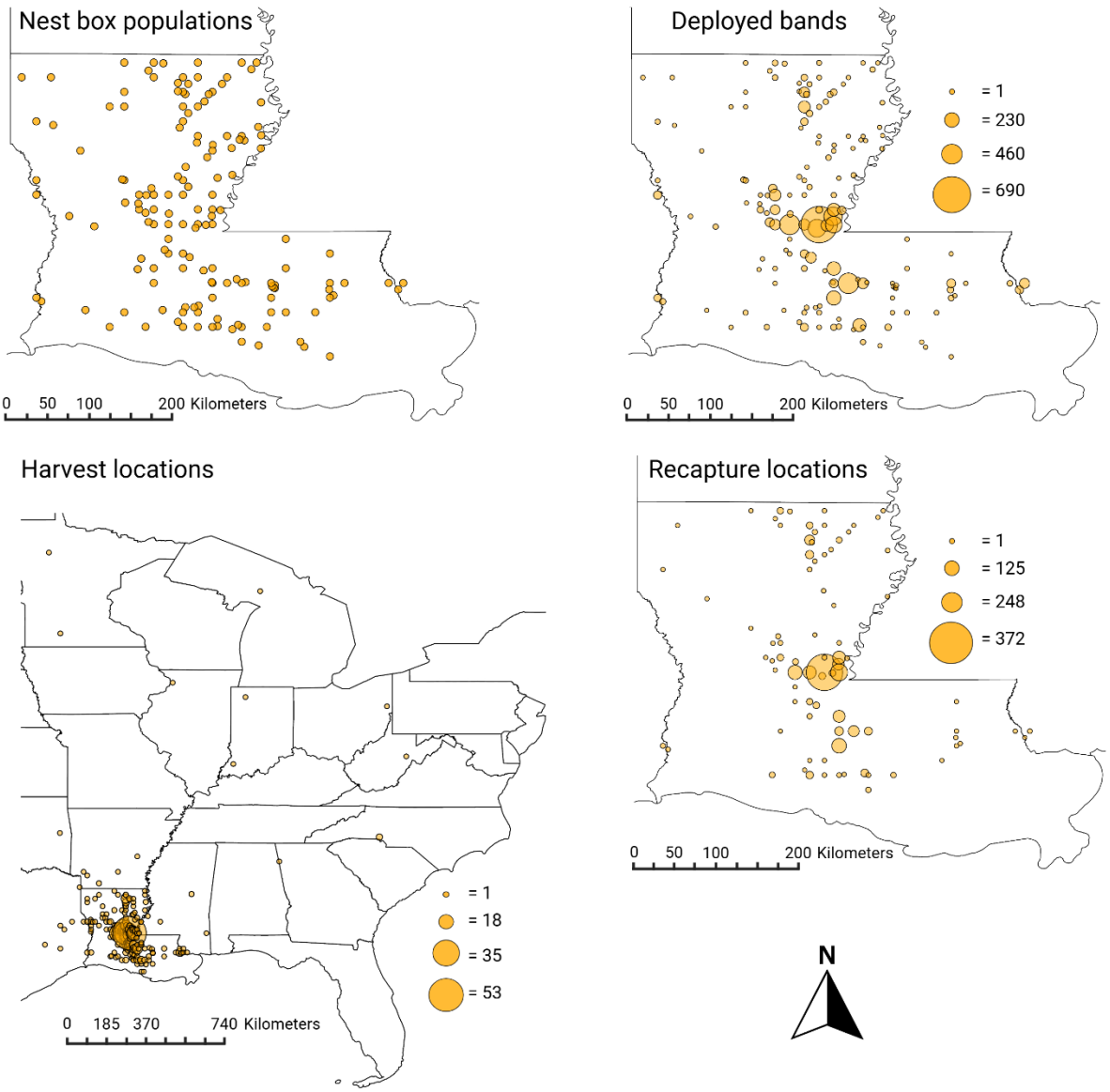

Figure 1. Louisiana maps showing the locations of nest box populations where adult female wood duck *Aix sponsa* bands were deployed, recovered by hunters (i.e., harvest locations), and recaptured by banders from 2001–2014. Survival and harvest rates were estimated using band data to investigate the impact of increasing the daily wood duck bag limit from two to three birds in 2008.

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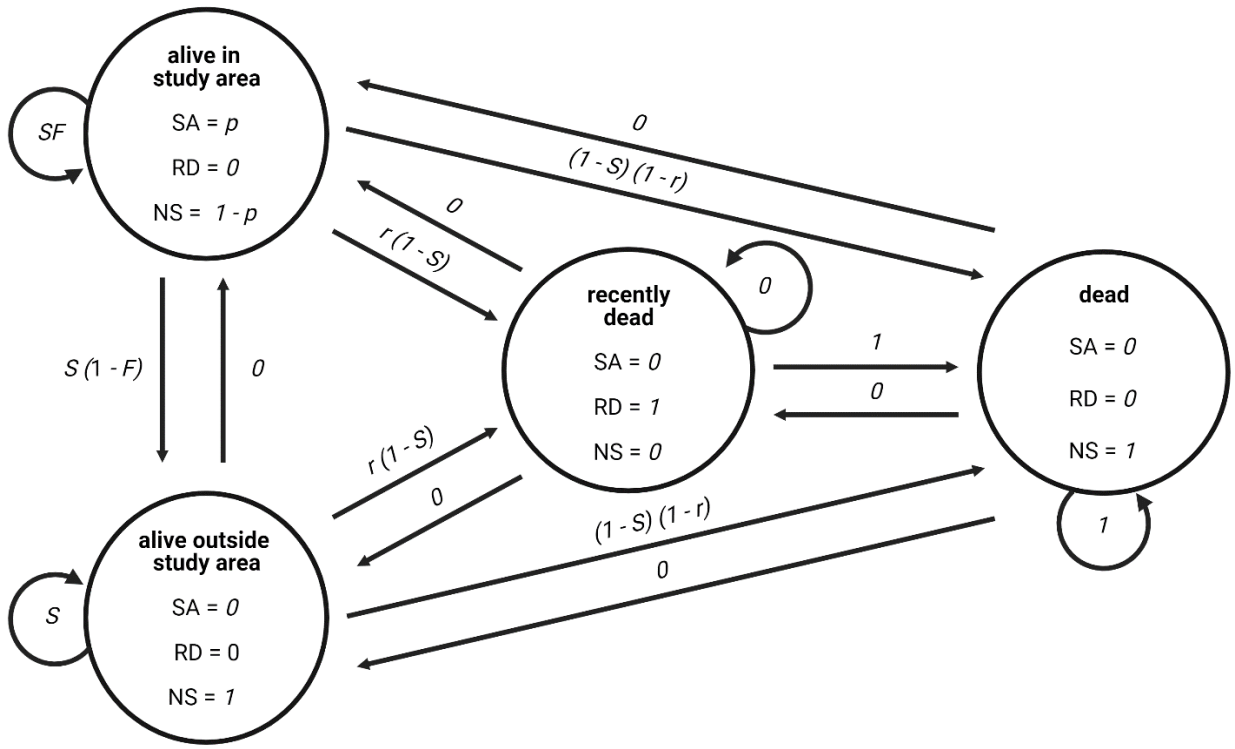

Figure 2. Conceptual diagram outlining the probabilities considered in state transition and observational matrices for a live-dead survival model for box-nesting wood ducks *Aix sponsa* in Louisiana. Alive in study area, alive outside study area, recently dead, and dead, were the states individuals could be in and made up the state transition matrix. The arrows represented the state transition probabilities where movement among states depended on survival (*S*), fidelity (*F*), and reported mortality (*r*) probabilities. The observational states were seen alive (SA), recovered dead (RD), and not seen or recovered (NS). The state individuals were observed in depended on recapture (*p*) probability (shown inside circles). The live-dead survival model was used to derive survival probabilities from 2001–2014 to investigate the consequences of increasing the daily wood duck bag limit from two to three birds in 2008.

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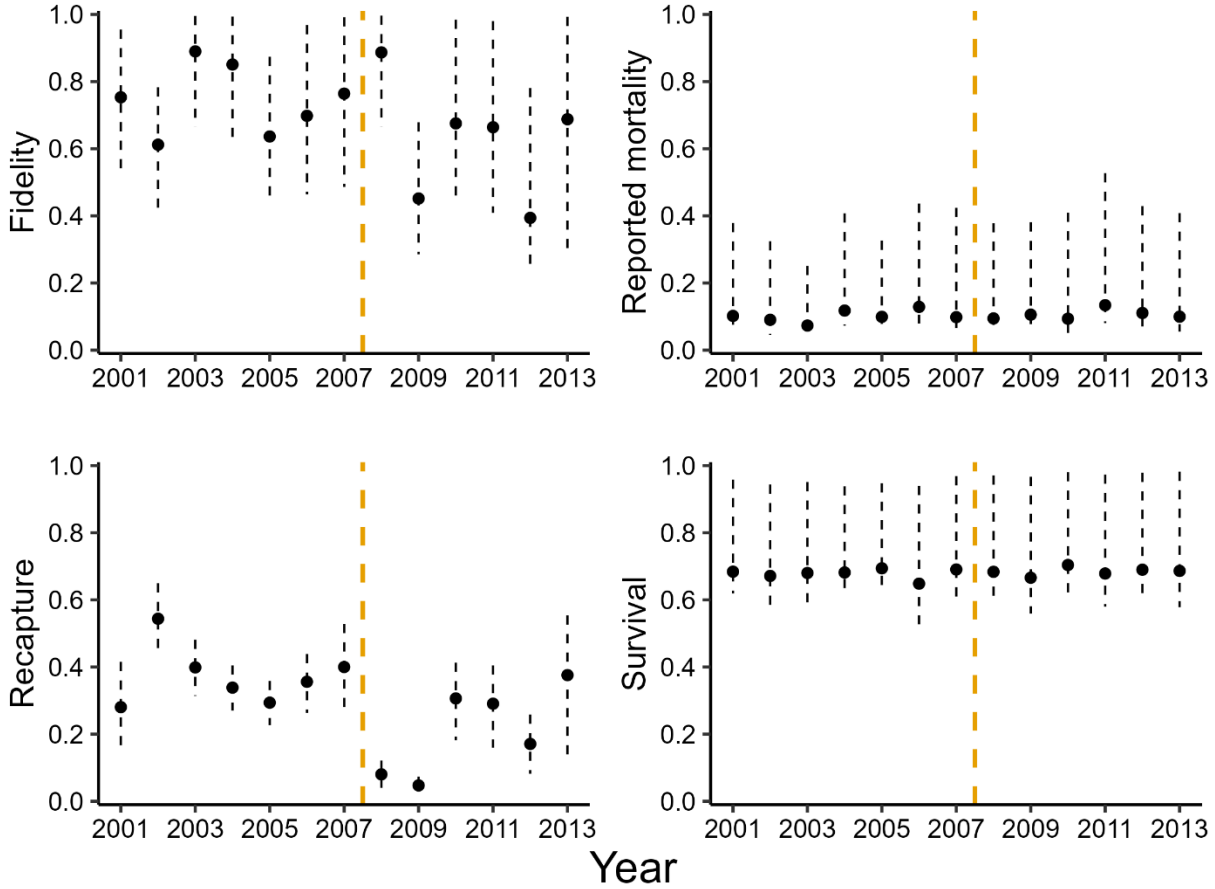

Figure 3. Probabilities of survival, fidelity, recapture, and reported mortality for female wood ducks *Aix sponsa* captured and banded from nest boxes in Louisiana from 2001−2014. The black dots represent posterior means and dashed lines show 95% credible intervals from a Bayesian live-dead analysis. Gold vertical dashed bars indicate when the daily wood duck bag limited increased from two to three.

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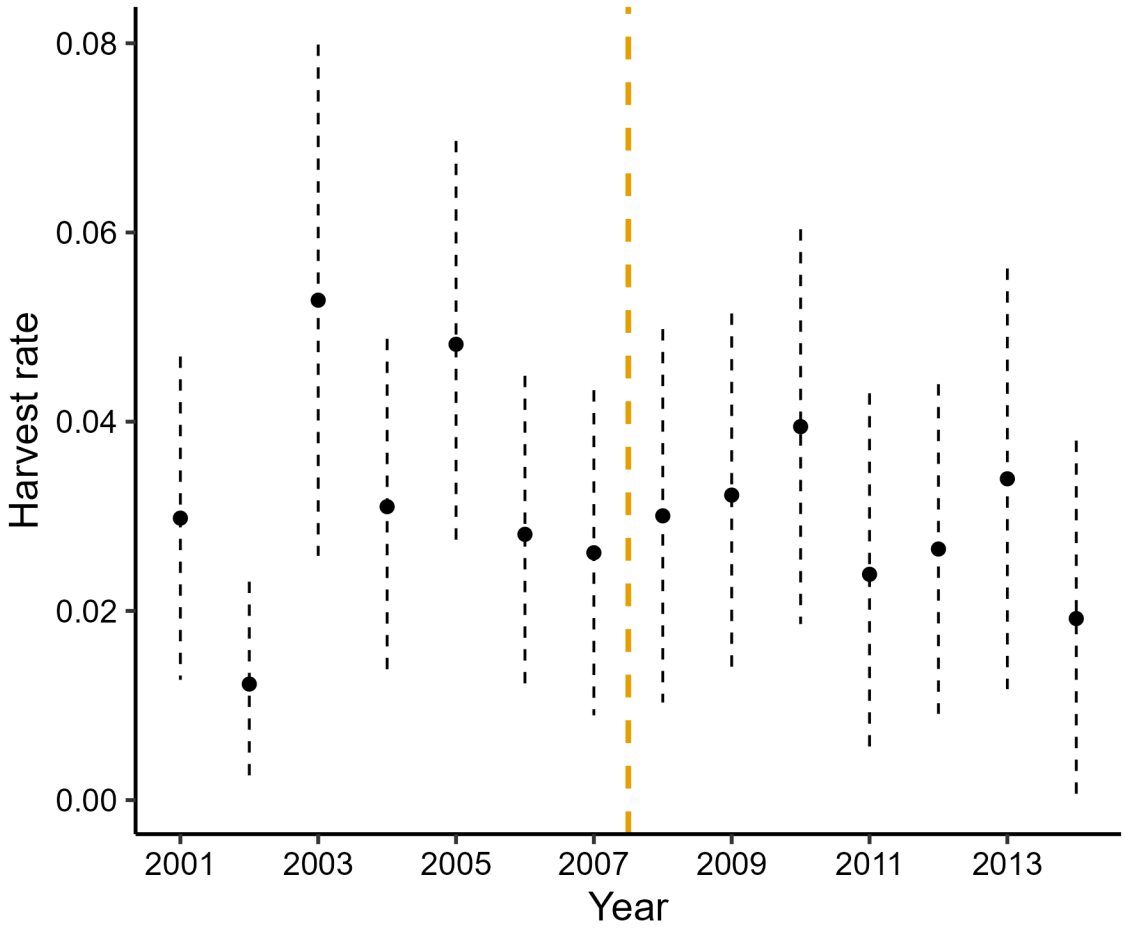

Figure 4. Harvest rate estimates for wood ducks *Aix sponsa* captured and banded from nest boxes in Louisiana from 2001−2014. The back dots are means and the dashed lines show 95% confidence intervals. Gold vertical dashed bars indicate when the daily wood duck bag limited increased from two to three.

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